

Project 4 — Lab 3

Silver Trust | n8n Automation | SIL-15: Automated Project Setup | March 2026

1. n8n Workflow

Workflow: SIL-15 — Automated Project Setup

Status: built, activated, and tested against mock webhook payload.

The workflow connects to Jira via the Jira SW Cloud credential. On execution it creates a Jira issue comment (the audit record), sends a Google Calendar invite, and sends Slack DMs — all triggered by a Jira allocation approval event.

2. Workflow Documentation

What it does

This workflow listens for a Jira webhook fired when consultants allocation status changes to "Approved". It fans out in parallel — sending one Google Calendar invite to the full team and individual Slack DMs to each assigned consultant — then posts a structured audit comment back to the originating Jira issue.

Workflow diagram

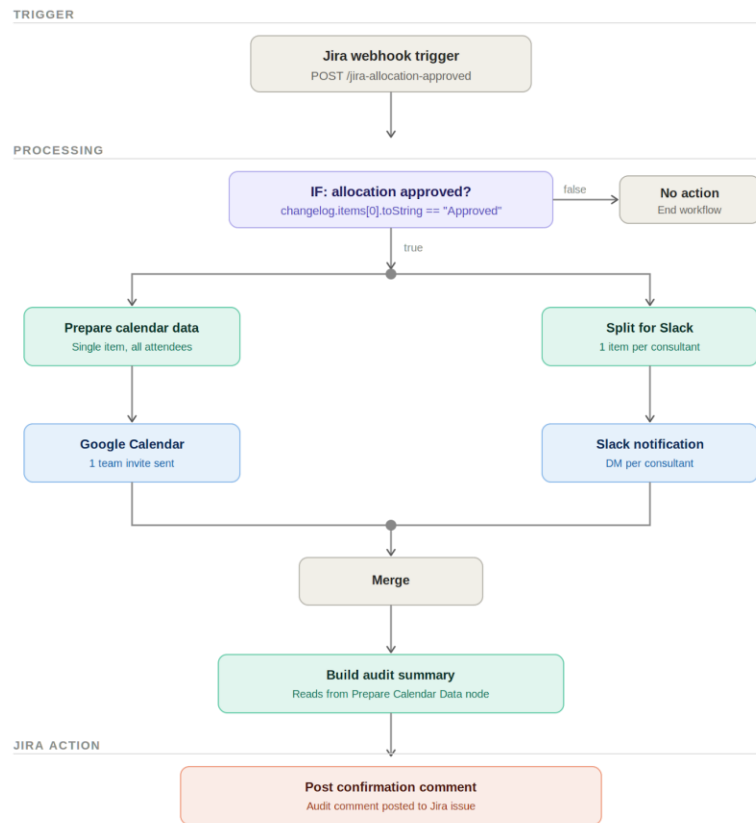


Figure 1 — SIL-15 n8n workflow: Trigger → Processing → Jira Action

Nodes and purposes

Node	Purpose	Key detail
Jira webhook trigger	Receives POST from Jira Automation	<i>Path: jira-allocation-approved</i>
IF: allocation approved?	Gates workflow on changelog value	<i>changelog.items[0].toString == Approved</i>
No action (NoOp)	Exits cleanly on false branch	<i>Non-approved statuses end here</i>
Prepare calendar data	Builds single calendar item	<i>Outputs attendeeEmails + consultants array</i>
Split for Slack	Splits consultants into N items	<i>One item per consultant</i>
Google Calendar	Sends one team invite	<i>Cred: CXa2ZaAzxZ1SmVol</i>

Slack notification	DMs each consultant individually	<i>Cred: 3j6374pjC4G3l6qQ</i>
Merge	Waits for both parallel branches	<i>Input 0: Calendar Input 1: Slack</i>
Build audit summary	Composes Jira comment	<i>Reads \$('Prepare Calendar Data').first()</i>
Post confirmation comment	Posts audit record to Jira issue	<i>Cred: dDawTjrp8sYwFXBj</i>

How to trigger / use

- Jira Automation rule fires a POST to the n8n webhook URL on allocation_status field change.
- For testing: open Jira Webhook Trigger1 in n8n → Listen for test event → Use mock data (SIL-15_mock_webhook_v3.json).
- Live trigger path: any Jira issue where changelog.items[0].toString transitions to "Approved".
- The workflow exits silently on any other value (NoOp branch).

3. Design Document

Architecture overview — Category 1 vs Category 2

Category 1 covers what n8n executes natively using deterministic logic. Category 2 covers capabilities requiring an LLM or ML model. SIL-15 is entirely Category 1; the AI recommendation stories (SIL-9, SIL-11, SIL-12, SIL-8) are Category 2.

Category 1 — n8n native (built / buildable)	Category 2 — LangChain / LLM (future)
Webhook trigger + Jira API (SIL-15 ✓)	Intelligent team recommendation scoring (SIL-9)
Google Calendar + Slack integration (SIL-15 ✓)	Semantic skill matching via embeddings (SIL-9)
Approval gate + audit comment (SIL-13)	Mentor/mentee pairing & alternative staffing (SIL-11)
Conflict detection — date range overlap (SIL-14)	Burnout-risk prediction from workload history (SIL-12)

Utilisation % alerting — threshold check (SIL-6)	Natural-language skill gap report (SIL-8)
Profile sync — form → Jira record (SIL-5)	Automatic project categorisation from free text

Implementation plan — what to build next

Phase	Workflow	Stories	Cat	Dependency
1 ✓	Automated Project Setup (webhook → Cal + Slack + Jira)	SIL-15	1	<i>Jira Automation webhook rule</i>
2	Approval gate — approve/reject with reason, triggers SIL-15	SIL-13	1	<i>SIL-15 complete; approval custom field</i>
2	Conflict detection — query Jira + Calendar, flag overlaps	SIL-14	1	<i>Calendar read access; allocation date fields</i>
3	Profile & utilisation sync — cron, 85% threshold alert	SIL-5, SIL-6	1	<i>Jira custom fields for consultant data</i>
4	Recommendation engine — LangChain scores + ranks consultants	SIL-9, SIL-11	2	<i>Phase 3 + LangChain service + vector store</i>
5	Burnout predictor + Skill gap report — LLM chains	SIL-12, SIL-8	2	<i>Phase 3 workload data; LLM API</i>

Technical constraints and recommendations

- No error branch — if Slack or Calendar fail mid-run, partial state is left; add a Try/Catch error workflow.
- Jira Automation rule not yet created — configure: Send webhook on allocation_status field change.
- Category 2 stories (SIL-9, 11, 12, 8) require a FastAPI microservice hosting LangChain chains + a vector store (Pinecone or pgvector); called from n8n via HTTP Request node.

- n8n Code nodes are sandboxed — libraries must be pre-installed; no runtime npm install.
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